UAS Avionics Enclosure

Requirements, Environments & Analysis Report — Revision D

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Project	Virtual environmental qualification of a machined avionics enclosure
Platform	7 in class long-range multirotor UAS, 6S LiPo
Material	6061-T6 aluminium, CNC machined
Tools	Fusion 360 (Nastran modal FEA), Python, Excel
Revision	D — random vibration, fatigue, thermal network, Phase 5 redesign, mass and envelope closed

1. Scope

Defines the design requirements and environmental load cases for an avionics enclosure housing the flight control, propulsion control and communications stack of a 7 in class multirotor UAS. Verification is by analysis only; no physical test article is produced. Every requirement traces to a manufacturer-published limit, a published military standard, or a platform-derived excitation condition.

- **In scope:** structural dynamics (modal, random vibration, fatigue), thermal (conduction, free and forced convection, radiation), mass, envelope.
- **Out of scope:** shock, EMI/EMC, ingress protection, altitude, humidity, physical test.

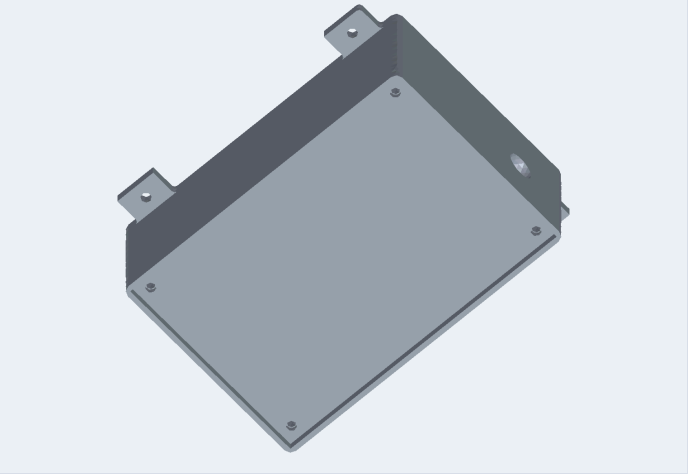


Figure 1 — Closed assembly, 93 × 63 × 33 mm.

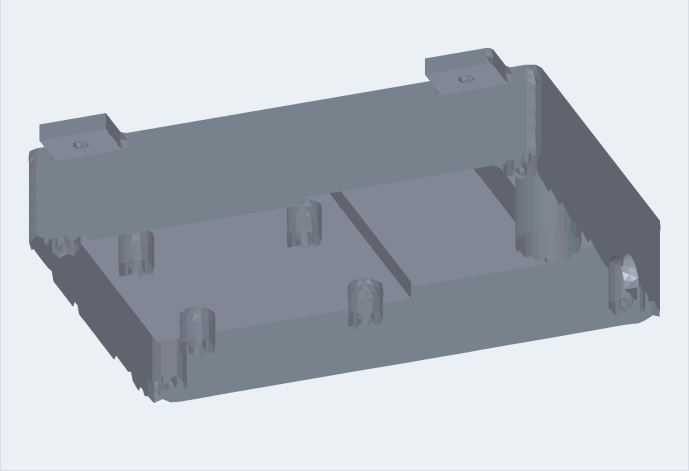


Figure 2 — Lid removed, showing the stack bay and the VTX / capacitor bay.

2. Component stack

ID	Component	Part	Mass (g)	Footprint (mm)	Height (mm)	Diss. (W)	Location
C1	Flight controller	Matek H743-SLIM	7.0	36 × 36	5.0	2.00 (A1)	Internal
C2	4-in-1 ESC	Holybro Tekko32 F4 65A	15.8	46 × 44	6.0	0.90 hover / 1.44 cruise	Internal
C3	Video transmitter	RushFPV Tank, 1.6 W class	12.0	37 × 24	6.7	**7.90**	Internal
C4	GPS / compass	Matek M10Q-5883	8.0	20 × 20	12.4	0.07	External
C5	RC receiver	Happymodel EP2 ELRS	0.4	10 × 10	6.0	0.20	External
C6	Bulk capacitor	Rubycon 35PK1000MEFCT810X20	2.75	⌀ 11.25	25.0	0.05	Internal
Internal heat load (C1 + C2 + C3)						**10.80 / 11.34**	**hover / cruise**

> **C4 and C5 are mounted externally.** An aluminium enclosure forms a Faraday cage; the M10Q-5883 uses an internal patch antenna and the EP2 an integrated SMD ceramic antenna. Neither would acquire signal if enclosed. Two thermal requirements were deleted as a consequence.

Temperature limits. The binding internal component allowable is **85 °C**, set by the C6 capacitor (Rubycon PK series, −40 to +85 °C). Rev C carried 105 °C, the generic electrolytic figure; this was corrected in Rev D. C1 carries a 40 °C figure interpreted as a maximum ambient (A4) alongside its ~100 °C component rating; the basis of the 40 °C figure was not found in Matek documentation and remains open.

C2 has no output BEC, so all 5 V loads draw from C1 and the conversion losses dissipate there. C2's larger 46 × 44 mm outline sets the minimum internal footprint. C3 runs from VBAT — the Tank family is specified for 7–36 V DC input (2–8S), which closes that open item.

3. Environments

3.1 Vibration — MIL-STD-810H Method 514.8

Profile: Annex E, Category 24 — General Minimum Integrity Exposure (Figure 514.8E-1). 20 Hz 0.04 g²/Hz → 1000 Hz 0.04 g²/Hz → −6 dB/oct → 2000 Hz 0.01 g²/Hz. Overall **7.7 g_{rms}** (integrated 7.69), 1 h per axis, 3 axes sequentially.

> **Why this profile.** No measured flight vibration data exists for this airframe, and no Annex D category represents a small electric multirotor — the rotary-wing categories are built around helicopter main- and tail-rotor sine tones. The minimum-integrity curve is the standard's own default when the environment is unknown, and its 20–2000 Hz band covers the prop-passing range. The standard states this profile is a minimum-integrity screen and cannot by itself qualify hardware. That caveat is carried forward: the results show the design is robust, not that it is qualified for a measured flight environment.

3.2 Platform excitation

T-Motor F90 2806.5, 1500 KV, 7 in 2-blade prop, 2,600–17,200 RPM. f = (RPM / 60) × N_{blades}.

Harmonic	1×	2×	3×	4×
At idle	87 Hz	173 Hz	260 Hz	347 Hz
At max	573 Hz	1,147 Hz	1,720 Hz	**2,293 Hz**

Excitation band: 87 to 2,293 Hz. A 1.5× separation factor on the top harmonic gives the 3,440 Hz target. A 2.0× factor would raise it to 4,587 Hz and the design would not close — recorded as a sensitivity rather than a failure, because the bonded-contact assumption (A6) already over-predicts stiffness.

3.3 Thermal conditions

Case	Freestream	Ambient	ESC current	ESC loss	Notes
Hover	≈ 0 m/s	35 °C	3.5 A/motor	0.90 W	Worst-case convection, low ESC load
Cruise	18 m/s	35 °C	6.0 A/motor	1.44 W	Good convection, higher I^2R loss

Current basis (A2, A8). A 1,000 g AUW needs 250 g thrust per motor at hover. At ~6.8 g/W for the F90 family on a 7 in prop, that is ~37 W, or 1.9 A per motor at 22.2 V. Cruise at 18 m/s adds parasite power $D \cdot V \approx 34$ W at 50% propulsive efficiency, about 2.7 A per motor. The analysis carries 3.5 A and 6.0 A — roughly 1.8× and 2.2× those estimates — to cover the missing 1500 KV thrust table and battery sag. **ESC loss model:** $P = 0.48\ \text{W} + 4 \times (I^2 \cdot 4\ \text{m}\Omega + 0.016\ \text{W/A} \cdot I)$.

4. Requirements and margins

ID	REQUIREMENT	PREDICTED	MARGIN	STATUS
REQ-STR-001	First mode > 3,440 Hz	3,984 Hz	+0.16	PASS
REQ-STR-002	No mode in 87–2,293 Hz	none in band	—	PASS
REQ-STR-003	3σ random vib vs yield	13.5 MPa	+16.8	PASS
REQ-STR-004	Fatigue, 1,000 h	D = 2.3e-7	life 4.3e9 h	PASS
REQ-THM-001	Internal air ≤ 40 °C	73.0 → 39.4 °C	+0.02	REDESIGN
REQ-THM-002	C1 ≤ 100 °C	89.7 °C	+0.11	PASS
REQ-THM-003	C2 FET ≤ 150 °C	79.4 °C	+0.89	PASS
REQ-THM-004	C3 case ≤ 90 °C	73.4 °C	+0.23	PASS
REQ-THM-005	C6 cap ≤ 85 °C	70.9 °C	+0.20	PASS
REQ-MAS-001	Mass ≤ 100 g	69.7 g	+0.43	PASS
REQ-ENV-001	≥ 1.0 mm clearance	1.00 mm	at limit	PASS
REQ-ENV-002	30.5 mm stack pattern	verified	—	PASS
REQ-ENV-003	Cable penetration	⌀ 8.0 mm	—	PASS

12 of 13 requirements close on the v1 design. REQ-THM-001 needs the Phase 5 redesign.

Figure 3 — All thirteen requirements with predicted values, margins and status.

Req ID	Condition	Predicted	Allowable	Margin	Status
REQ-STR-001	Modal	3,984 Hz	3,440 Hz	**+0.16**	**PASS**
REQ-STR-002	Modal	f_1 = 3,984 Hz	87–2,293 Hz	**No mode in band**	**PASS**
REQ-STR-003	Random vib, Q = $\sqrt{f_n}$	13.5 MPa (tab)	241 MPa	**+16.8**	**PASS**
REQ-STR-004	Fatigue, 1,000 h	D = 2.3e−7	D < 1	**life 4.3e9 h**	**PASS**
REQ-THM-001	Hover, anodised, v1	**73.0 °C**	40 °C	**−0.45**	**FAIL**
REQ-THM-001	Cruise, anodised, v1	**48.4 °C**	40 °C	**−0.17**	**FAIL**
REQ-THM-001	Hover, Phase 5 R4	39.4 °C	40 °C	**+0.02**	**PASS (marginal)**
REQ-THM-001	Cruise, Phase 5 R4	37.4 °C	40 °C	**+0.07**	**PASS**
REQ-THM-002	Hover, anodised, v1	89.7 °C	100 °C	**+0.11**	**PASS**
REQ-THM-003	Hover, anodised, v1	79.4 °C	150 °C	**+0.89**	**PASS**
REQ-THM-004	Hover, anodised, v1	73.4 °C	90 °C	**+0.23**	**PASS**
REQ-THM-005	Hover, anodised, v1	70.9 °C	85 °C	**+0.20**	**PASS**
REQ-MAS-001	v1 (base + lid + 8 M3)	69.7 g	100 g	**+0.43**	**PASS**
REQ-MAS-001	Phase 5 R4 (fins + 2 fans)	99.3 g	100 g	**+0.01**	**PASS (marginal)**
REQ-ENV-001	CAD	0 interferences, min 1.00 mm	1.0 mm	0.00	**PASS (at limit)**
REQ-ENV-002	CAD	30.5 × 30.5 mm, 4 × M3	30.5 × 30.5 mm	—	**PASS**
REQ-ENV-003	CAD	⌀ 8.0 mm, +X wall	Present	—	**PASS**

> **Bare aluminium is not acceptable.** At hover, bare v1 fails REQ-THM-002 (110.6 °C), REQ-THM-004 (90.9 °C) and REQ-THM-005 (88.9 °C). An external **and** internal anodise (ε ≈ 0.85) is a design requirement, not a finish option.

5. Modal results

Ten modes in Fusion 360 (Nastran). Fixed on the four mounting tab underside faces, bonded at all interfaces (A6). No mode falls in the excitation band; the first mode clears the fourth harmonic at maximum RPM by a factor of 1.74.

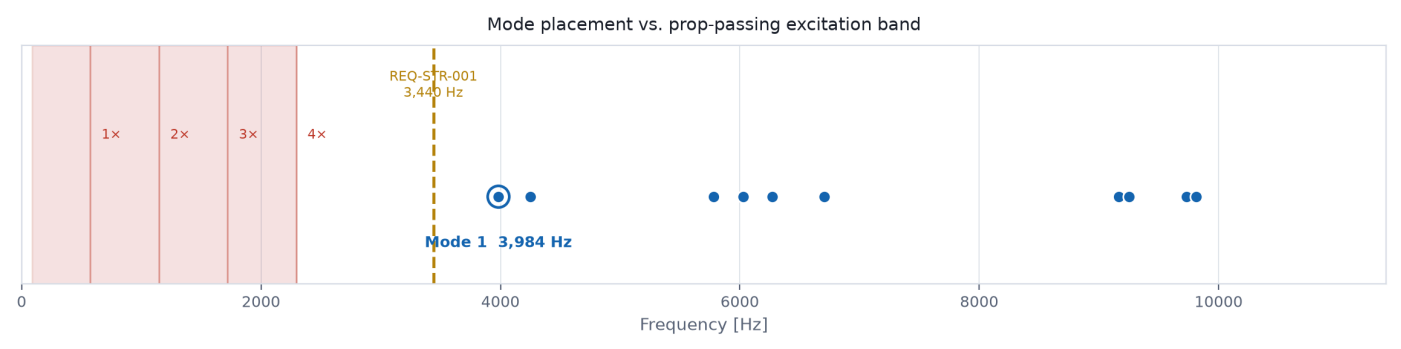


Figure 4 — Mode placement against the excitation band.

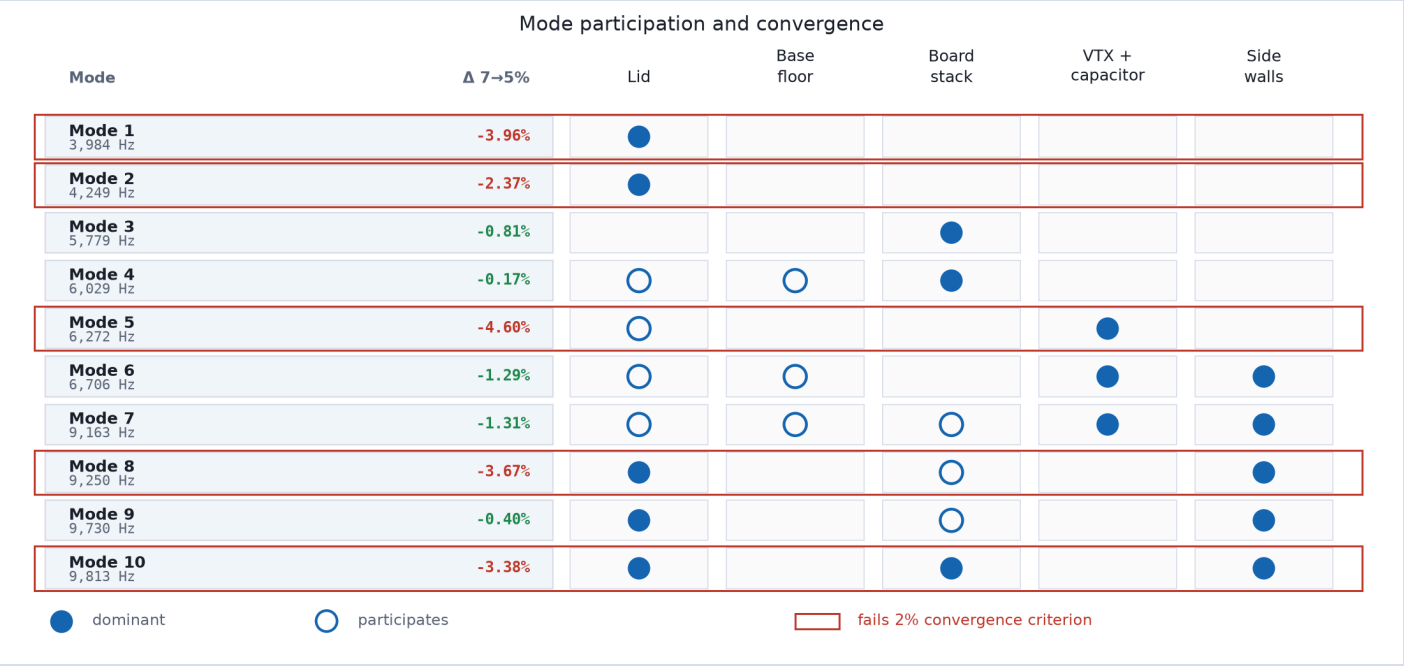


Figure 5 — Which parts move in each mode, with convergence status. Every mode failing the 2% criterion has the lid or capacitor dominant.

Mode	Freq, 10% mesh	Description
1	4286.5 Hz	Lid drumming, first bending. Centre antinode decaying to zero at the four corner bolts. Base and stack essentially rigid.
2	4690.0 Hz	Second lid mode. Lid deforms; base floor does not participate.
3	5882.3 Hz	Board stack responds about Y. VTX and capacitor stationary.
4	6071.6 Hz	Small coupled lid and floor bending. Stack bends about X.
5	6955.8 Hz	Capacitor and partial VTX response. Lid deforms out of plane over the stack.
6	7130.9 Hz	Capacitor-side wall deforms inward with the VTX. Lid and floor in opposition.
7	9520.8 Hz	Capacitor fully coupled into the adjacent wall. Lid and floor anti-phase diagonal.
8	9847.4 Hz	Coupled lid and long-wall response. Capacitor nodal.
9	10081.1 Hz	Long-wall out-of-plane bending. Two lid antinodes either side of the cable penetration.
10	10591.3 Hz	Three-antinode side-wall mode. Lid bows up while stack bows down.

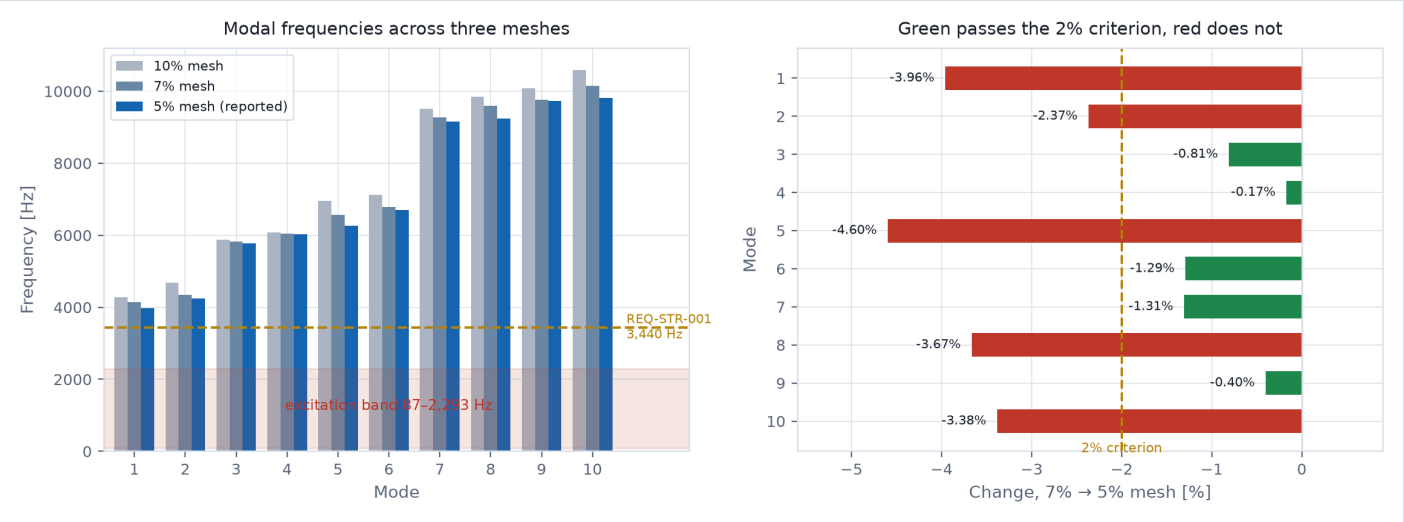


Figure 6 — Three-mesh convergence study.

Mode 1 does not meet the 2% criterion. The shift grew from 3.23% to 3.96% across successive refinements, and Richardson extrapolation returns a negative observed order. Modes 3 and 4 converge cleanly, and both are dominated by the single-body base. The poorly converging modes (1, 2, 5, 8, 10) all involve the lid or capacitor, which depend on bonded contact between separately meshed bodies. The reported 3,984 Hz is the lowest of three meshes and therefore the most conservative available. **Sensitivity:** at ~4% per refinement, REQ-STR-001 would still hold for about three more halvings (3,984 → 3,440 Hz is a 13.6% drop), and the random-vibration margin is insensitive to f_1 because Miles' response scales as $\sqrt{f_n}$.

6. Random vibration and fatigue

f_1 lies above the 2 kHz profile limit, so the PSD is extrapolated at −6 dB/oct to f_1 (0.00252 g²/Hz). The profile has no energy there, so this is conservative. The response is taken as the larger of Miles' equation and the quasi-static input (7.69 g_{rms}).

Case	g_{rms}	3σ (g)	Lid plate σ (MPa)	Tab root σ, K_t 2.5 (MPa)	MS vs 241 MPa
Q = 10 (nominal)	12.6	37.7	0.44	5.39	**+43.7**
Q = $\sqrt{f_n}$ = 63 (Steinberg bound)	31.6	94.7	1.11	13.54	**+16.8**

- **Lid:** Roark Table 11.4 case 1a, simply supported 90.5 × 60.5 × 4.0 mm plate under 3σ inertial pressure.
- **Tabs:** 105 g assembly on four 12 × 3 mm cantilevers, 4 mm moment arm, K_t = 2.5.
- **Fatigue** (governing tab, Q = $\sqrt{f_n}$): 1σ = 4.51 MPa. Steinberg three-band cycle counts over 1,000 h at 3,984 Hz give Miner damage 2.3 × 10^{-7}. The 3σ peak of 13.5 MPa is below the 96.5 MPa endurance strength at 5 × 10^8 cycles, so life is effectively unlimited.
- **Not covered:** bolt preload loss and fretting at the bonded interfaces (A6), and board-level solder fatigue (A7).

7. Thermal network and Phase 5 redesign

Six-node steady-state network (shell, internal air, C1, C2, C3, C6) with geometry from the Fusion model. It agrees with the earlier lumped scoping calculation — shell 68.8 vs 69.3 °C at hover anodised, well inside the 30% cross-check criterion. Key conductances: C3 → shell 0.59 K/W (0.5 mm, 3 W/m·K gap pad on the 666 mm² machined pad plus floor spreading); C2 → shell 3.94 K/W (four M3 bosses); C1 → C2 8.21 K/W (four aluminium standoffs). Internal h = 3 W/m²·K still air, 15 W/m²·K with fans.

Case	Q (W)	Shell	Air	C1	C2	FET j	C3	C6
Hover, bare Al	10.85	86.2	**91.1**	**110.6**	96.2	98.3	**90.9**	**88.9**
Hover, anodised	10.85	68.8	**73.0**	89.7	77.3	79.4	73.4	70.9
Cruise, bare Al	11.39	44.4	**49.8**	70.8	56.4	61.2	49.0	47.2
Cruise, anodised	11.39	43.6	**48.4**	66.7	54.2	59.0	48.2	45.8

Hover governs every requirement: the low hover ESC load is outweighed by the loss of forced convection. REQ-THM-001 allows 5 K of air rise for ~11 W, needing an air-to-ambient conductance of ~2.2 W/K. The sealed v1 box provides ~0.32 W/K at hover, and closing that with surface area alone would take ~6.9× the external area. The redesign therefore stacks four measures.

Option	Change	Mass Δ
R1	Anodise inside and out (baseline above)	0
R2	9 longitudinal lid fins, 1.0 mm × 10 mm tall, 7 mm pitch, machined integral (+0.0167 m²)	+22.6 g
R3	VTX operated at the 800 mW level (published draw 380 mA @ 12 V = 4.56 W)	0
R4	Two 25 mm 5 V fans, push-pull through filtered ducts (~0.47 L/s each, 0.8 W motor heat)	+7 g

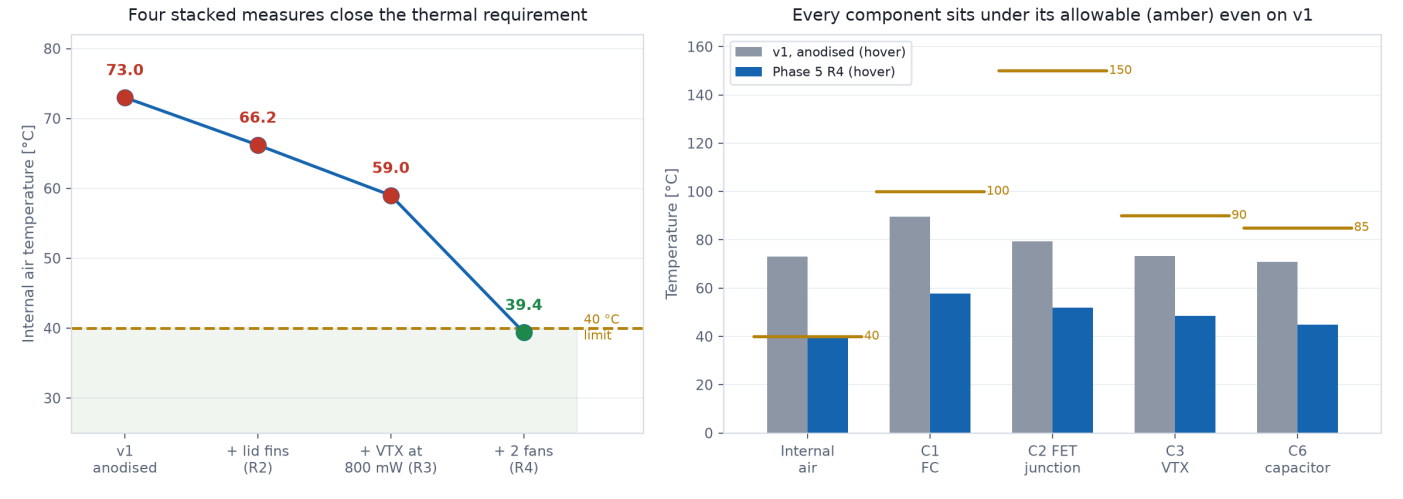


Figure 7 — Left: internal air temperature through the four stacked measures at hover. Right: component temperatures on v1 and Phase 5 R4 against their allowables.

Case	Q (W)	Shell	Air	C1	C2	FET j	C3	C6
Hover, R1 no TIM	10.85	68.8	73.4	89.7	77.3	79.4	79.1	70.9
Hover, R2 + fins	10.85	62.0	66.2	83.1	70.6	72.7	66.6	64.1
Hover, R3 + VTX 800 mW	7.51	54.8	59.0	76.1	63.5	65.6	57.5	56.9
Hover, R4 + 2 fans	8.31	45.9	**39.4**	57.7	49.9	52.0	48.5	44.8
Cruise, R1 no TIM	11.39	43.6	48.8	66.8	54.2	59.0	54.0	45.9
Cruise, R2 + fins	11.39	40.7	45.5	63.9	51.3	56.1	45.3	42.9
Cruise, R3 + VTX 800 mW	8.05	39.0	43.7	62.3	49.7	54.5	41.7	41.3
Cruise, R4 + 2 fans	8.85	38.1	**37.4**	53.5	45.2	50.0	40.8	39.1

Findings

1. All component requirements (REQ-THM-002 to 005) close on the anodised v1 **with a gap pad under the VTX**. Without the pad, C3 rises 5.7 K at hover; the gap pad is a design requirement.
2. REQ-THM-001 closes only with the full R2–R4 stack, and at hover only marginally (39.4 °C, 0.6 K margin). It is not robust to the network assumptions — internal h, fan delivered flow, or A4.
3. R3 costs video range (1.6 W → 800 mW, roughly −30% range at equal link margin). R4 breaks the sealed-box concept; ingress protection is out of scope, but the inlet needs a filter.

> **Recommendation: close open item A4 before committing to R2–R4.** If the 40 °C figure is not a real C1 ambient rating, REQ-THM-001 should be restated to the most limiting internal-component ambient. The STM32H743 is rated −40 to +85 °C ambient and the C6 rating is 85 °C. On that basis the anodised v1 with a gap pad passes with MS ≥ +0.11 at hover and needs no fans, fins or VTX derating. If the 40 °C is confirmed, adopt R2 + R3 + R4 as enclosure_v2.

8. Mass and envelope

Item	Value	Item	Value
Base (Body60)	54.26 g	Interferences (10 bodies)	0
Lid (Body61)	11.87 g	Min clearance (C6–lid)	1.00 mm
Base + lid	66.13 g	C6–C3 / C2–C3 / C1–lid	1.38 / 2.00 / 3.00 mm
+ 8 × M3 steel screws	69.7 g	Stack pattern	30.5 × 30.5 mm
Assembly incl. stand-ins	101.34 g	Cable penetration	⌀ 8.0 mm, +X wall

The C1 and C2 stand-in masses (8.43 g, 10.93 g) differ from published values (7.0 g, 15.8 g). Net assembly error is −3.4 g (−3%), which raises the modal frequencies slightly and is tracked as an open item.

9. Assumptions log

#	Assumption	Justification	Impact if wrong
A1	C1 dissipation 2.00 W	Published static 1.0 W, rounded up to bound BEC losses	**Over-predicts ~0.6 W. Conservative.**
A2	C2 0.90 W hover, 1.44 W cruise	Loss model calibrated to 0.90 W at 3.5 A	**Conduction scales with I^2. C2 is 8–13% of load; low sensitivity.**
A3	Ambient 35 °C	Hot-day condition	**Offsets every temperature 1:1**
A4	C1 40 °C read as max ambient	Distinct from ~100 °C component rating. No Matek source found.	**Drives the entire Phase 5 redesign.**
A5	Standoffs modelled as aluminium	Stiffer than nylon	**Nylon lowers stack modes and raises C1 temperature. Non-conservative.**
A6	All interfaces bonded	Fully preloaded joints	**Over-predicts stiffness. NON-CONSERVATIVE for REQ-STR-001.**
A7	Components as homogeneous solids	Correct mass and envelope	**Board-level modes and solder fatigue not represented**
A8	AUW 1,000 g; currents from F90 KV1300 efficiency data	No published 1500 KV / 7 in thrust table found	**Carried at ~2× the estimate; low thermal sensitivity**
A9	Service life 1,000 flight hours	Typical for a long-range FPV platform	**Damage 2.3e−7 per 1,000 h; insensitive**
A10	Mass budget 100 g (10% of AUW)	Allocation, not a sourced limit	**v1 has 30 g margin; R4 has 0.7 g**
A11	Network conductances (gap pad 3 W/m·K, 15 K/W per PCB-to-boss joint, internal h 3 W/m²·K)	Handbook values	**C1 and C2 most sensitive. Verify with P-THM-1.**
A12	MIL-STD-810H Cat. 24 minimum-integrity profile	No measured flight data	**Screening-level only; not a qualification environment**
A13	Response Q = $\sqrt{f_n} \approx 63$ (Steinberg bound)	Conservative vs typical Q = 10–20 for bolted Al	**Over-predicts stress ~2.5×. Conservative.**

10. Fusion procedures and open items

The Fusion adsk.sim scripting API returns no simulation object in the installed build, and its documentation limits scripted solves to Linear Static Stress. The following studies are therefore run interactively and recorded against the hand calculations.

- **P-STR-1 (done):** Modal, 10 modes, fixed tab undersides, bonded contacts, 10/7/5% mesh.
- **P-STR-2 (to run):** Linear Static Stress, gravity load of 94.7 g applied separately in X, Y, Z. Compare max von Mises in tabs and lid against 13.5 / 1.1 MPa.
- **P-THM-1 (to run):** Thermal study, hover anodised. Compare shell and C3 against 68.8 / 73.4 °C; divergence above 30% needs explanation.
- **P-THM-2 (to run):** As P-THM-1 at cruise conditions.

Open items. Confirm the basis of the C1 40 °C figure (decides whether R2–R4 are needed); confirm C1 board revision; obtain the F90 1500 KV / 7 in thrust table to replace A8; obtain the Rubycon PK datasheet to confirm 85 °C; correct the C1 / C2 stand-in masses in CAD and re-run modal; replace the bonded lid contact with a bolted/separation contact to investigate mode 1 non-convergence; run P-STR-2, P-THM-1 and P-THM-2.

11. Revision history

Rev	Change
A	Initial requirements
B	Modal results incorporated
C	C4/C5 moved external; mesh convergence; motor/prop and separation factor set
D	Vibration profile selected; REQ-STR-003/004 closed by analysis; thermal network added; ESC currents stated; C6 rating corrected 105 → 85 °C; Phase 5 redesign sized; mass, envelope, stack pattern and penetration verified from CAD; Fusion procedures defined